# From OpenAI to Open Source AI

*Navigating between Commercial Ownership and Collaborative Openness*



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Open Source Expert
Senior Architect at Worldline

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Who are we?

# We design payments technology that powers the growth of millions of businesses around the world.



7000+ engineers
in over 40 countries



Managing 43+ billion
transactions per year



€250M spent in R&D
every year



Handling 150+
payment methods

# The **early days** of LLMs

**From rule-based and simpler statistical models to LLMs**



**2010’s**

Word embeddings
Word2Vec, GloVe



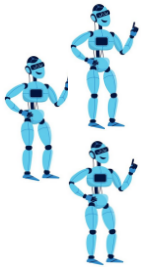
**2017-2018**

“*Attention is All You Need*"
Transformers



**2020’s**

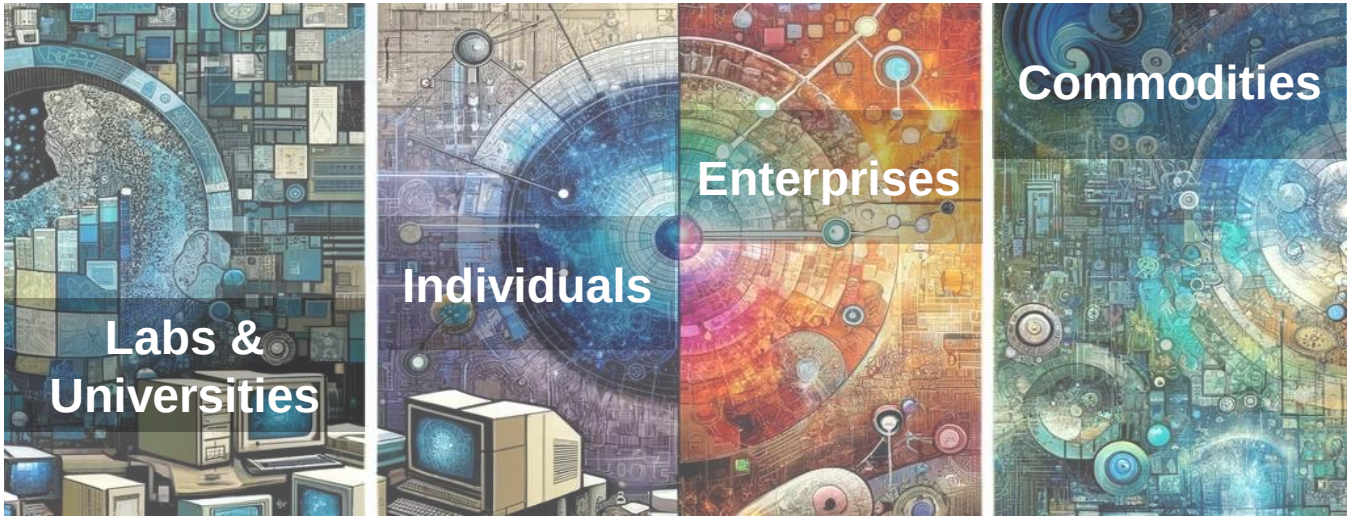
GenAI, ChatGPT
Responsibility concerns



**Tomorrow?**

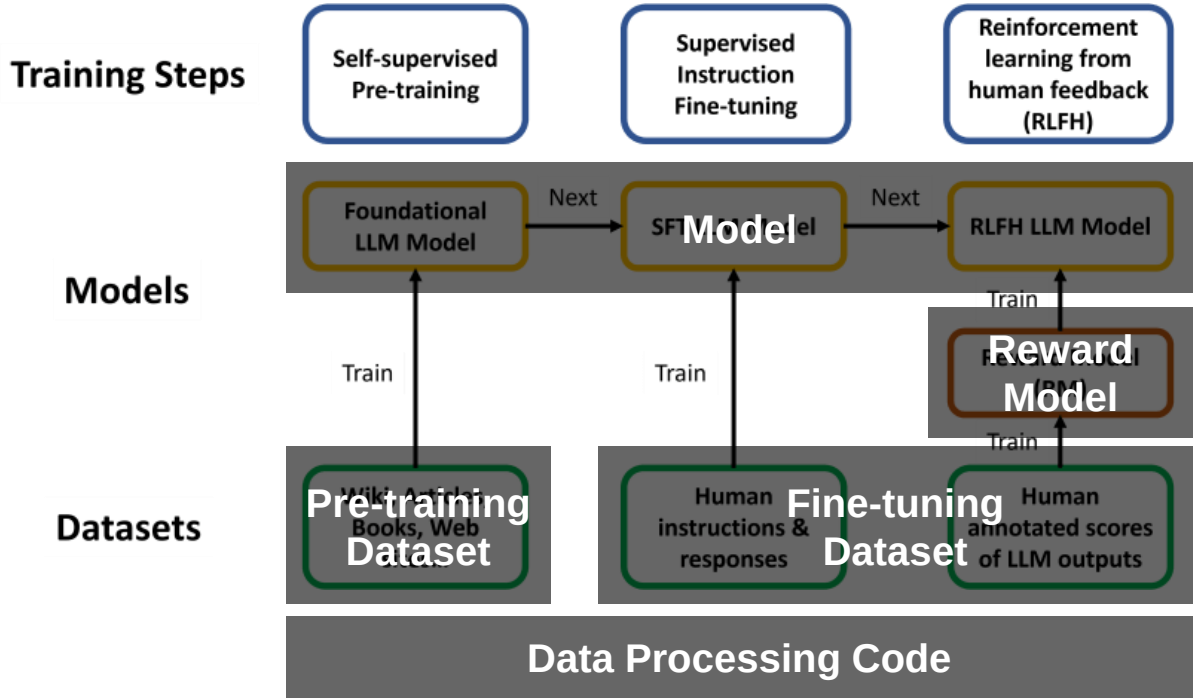
Small Language Models
Mobile, Agents & LAMs

# GenAI is having its **Linux Moment**




- Just like open source and Internet, bust much faster!
- Dynamics between collaborative openness and commercial ownership
- Need of clarity on licenses

# Defining **Openness** of a Model



# Defining **Openness** of a Model

| Score | Level | Description | Model (weights) | Pre-training Dataset | Fine-tuning Dataset | Reward model | Data Processing Code |
|---|---|---|---|---|---|---|---|
| 0 | **Closed** | **No access** to any public information, data or asset | | | | | |
| 1 | **Published research only** | **Research papers(s)** published but with no more information, data or asset | | | | | |
| 2 | **Restricted access** | Access to asset is possible only with **special agreement** (commercial, research…) | | | | | |
| 3 | **Open with limitations** | Access and reuse of asset is possible but with certain **limitations on usage** | | | | | |
| 4 | **Totally open** | Access and reuse of asset is possible **without restriction on usage** (ex. open source license) | | | | | |

# **Market-Leading Player**: OpenAI

**Deviation from original vision of research transparency & openness**




GPT-1 & 2

| Non/For-profit (US) | | |
|---|---|---|
| **Component** | **Score** | **Level description** |
| **Model** | 4 | Totally open |
| **Dataset** | 1 | Published research only |
| **Code** | 1 | Published research only |

→

GPT-3.x & 4.x/o

| | |
|---|---|
| 0 | Closed |

ChatGPT

research paper only

# **Market-Leading Player**: OpenAI

**Deviation from original vision of research transparency & openness**

GPT-1 & 2

GPT-3.x & 4.x/o

| Non/For-profit (US) | | |
|---|---|---|
| | | only |
| Code | 1 | Published research only |

*You may not: […] Use **Output** to develop models that compete with OpenAI.*

| | |
|---|---|
| 0 | Closed |

ChatGPT

research paper only

No training of **other** commercial LLMs

# **Market-Leading Player**: Google

## Transition from open research to a pragmatic approach

Google BERT

| Enterprise (US) | | |
|---|---|---|
| **Component** | **Score** | **Level description** |
| **Model** | 4 | Totally open |
| **Dataset** | 2 | Restricted access |
| **Code** | 4 | Totally open |

→

PaLM 2

Gemini

| | |
|---|---|
| 1 | Published research only |
| 1 | Published research only |
| 0 | Closed |

↔

Gemma

| | |
|---|---|
| 3 | Open with limitations |
| 1 | Published research only |
| 4 | Toolchain available |

# **Market-Leading Player**: Google

**Transition from open research to a pragmatic approach**

Google BERT

PaLM 2

Gemma

| Component | | |
|---|---|---|
| Model | | |
| Dataset | | |
| Code | 4 | Totally open |

| | |
|---|---|
| 3 | Open with limitations |
| 1 | Published research only |
| 4 | Toolchain available |

*You may **not use nor allow others to use** Gemma or Model Derivatives to: [illegals activities, unlicensed practices of profession, abuse, security bypass and promotion of hatred, abuse, violence, monitoring people without consent, misinformation/defamation, automate decisions concerning human rights and well-being, etc.]*



**Responsible AI** contradicts Open Source Definition

# Other **Big Players**

**Catching up and making their mark in the GenAI Gold Rush**

Create their own (open) models



















Partner for Infrastructure (inference and training)

# **Market-Leading Player**: Meta

## Journey to openness

RoBERTa

| Enterprise (US) | | |
|---|---|---|
| **Component** | **Score** | **Level description** |
| **Model** | 4 | Totally open |
| **Dataset** | 3 | Open with limitations |
| **Code** | 4 | Totally open |

→

LLaMA by Meta

| | |
|---|---|
| 3 | Open with limitations |
| 1 | Published research only |
| 1 | Published research only |

# **Market-Leading Player**: Meta

**Journey to openness**

RoBERTa

LLaMA

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**Restriction on usage**: license for platforms with 700+ M users

# **Market-Leading Player**: Meta

**Journey to openness**

RoBERTa

LLaMA

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LLaMA 3 now more restrictive on **redistribution and reuse**

# **Llama 2 offspring's**: Alpaca and Vicuna

**Fine-tuned models from Llama 2 by universities**







| Research (US) | | |
|---|---|---|
| **Component** | **Score** | **Level description** |
| **Model** | 3 | Open with limitations |
| **Pre-training Dataset** | 1 | Published research only |
| **Fine-tuning Dataset** | 2 | Research use only |
| **Code** | 4 | Under Apache 2 license |



Restrictions from **both** Llama 2 and OpenAI (ShareGPT)

# Collaborative **foundational** LLMs

| | Non-profit (US) | | Research (UAE) | | Research (EU) | | Research (US) | | Enterprise (FR) | |
|---|---|---|---|---|---|---|---|---|---|---|
| | EleutherAI GPT-J | | Falcon | | BLOOM | | OpenLLaMa | | Mistral | |
| **Model** | 4 | Access and reuse without restriction | 3 | Open with limitations | 3 | Open RAIL license | 4 | Access and reuse without restriction | 4 | Access and reuse without restriction |
| **Dataset** | 3 | Open with limitations | 4 | Access and reuse without restriction | 3 | Open with limitations | 4 | Access and reuse without restriction | 0 | No public information or access |
| **Code** | 4 | Completely open | 1 | General instructions | 4 | Completely open | 1 | Just examples | 4 | Completely open |

**Dataset** fuzziness: *please refer to the specific license depending on the subset you use*
Notion of **responsible** usage

# Collaborative **foundational** LLMs

| | Non-profit (US) | Research (UAE) | Research (EU) | Research (US) | Enterprise (FR) |
|---|---|---|---|---|---|
| | EleutherAI GPT-J | Falcon | BLOOM | OpenLLaMa | Mistral |
| Model | Access and | | | Access and | Access and |
| Dataset | | | | | public |
| Code | | | | | completely |

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**Modified** open source licenses

# Mistral AI's French sauce

**Navigating both open and close waters**

MISTRAL AI_

## Mix of AI Models

- Mixture-of-Experts (SMoE): Mixtral 8x7B, 8x22B
- Foundational and fine-tuned models



## Mix of Business Models & Licenses

- "Open Source" models, mistral-finetune SDK
- Commercial: optimized Small, Large & Embed Models
- Sustainable openness: new non-production license for codestral



Just like with Open Source, rise of **Community VS Enterprise**

# Mistral AI's French sauce

## Navigation both open and close waters

**Mix**

- M
- F

**Mix**

- "C
- Commercial: optimized Small, Large & Embed Models
- Sustainable openness: new non-production license for codestral

> ***MNPL - 3.2. Usage Limitation***
> *- You shall only use the Mistral Models and Derivatives (whether or not created by Mistral AI) for testing, research, Personal, or evaluation purposes in Non-Production Environments;*
> *- Subject to the foregoing, You shall not supply the Mistral Models or Derivatives in the course of a commercial activity, whether in return for payment or free of charge, in any medium or form, including but not limited to through a hosted or managed service (e.g. SaaS, cloud instances, etc.), or behind a software layer.*

MISTRAL AI_





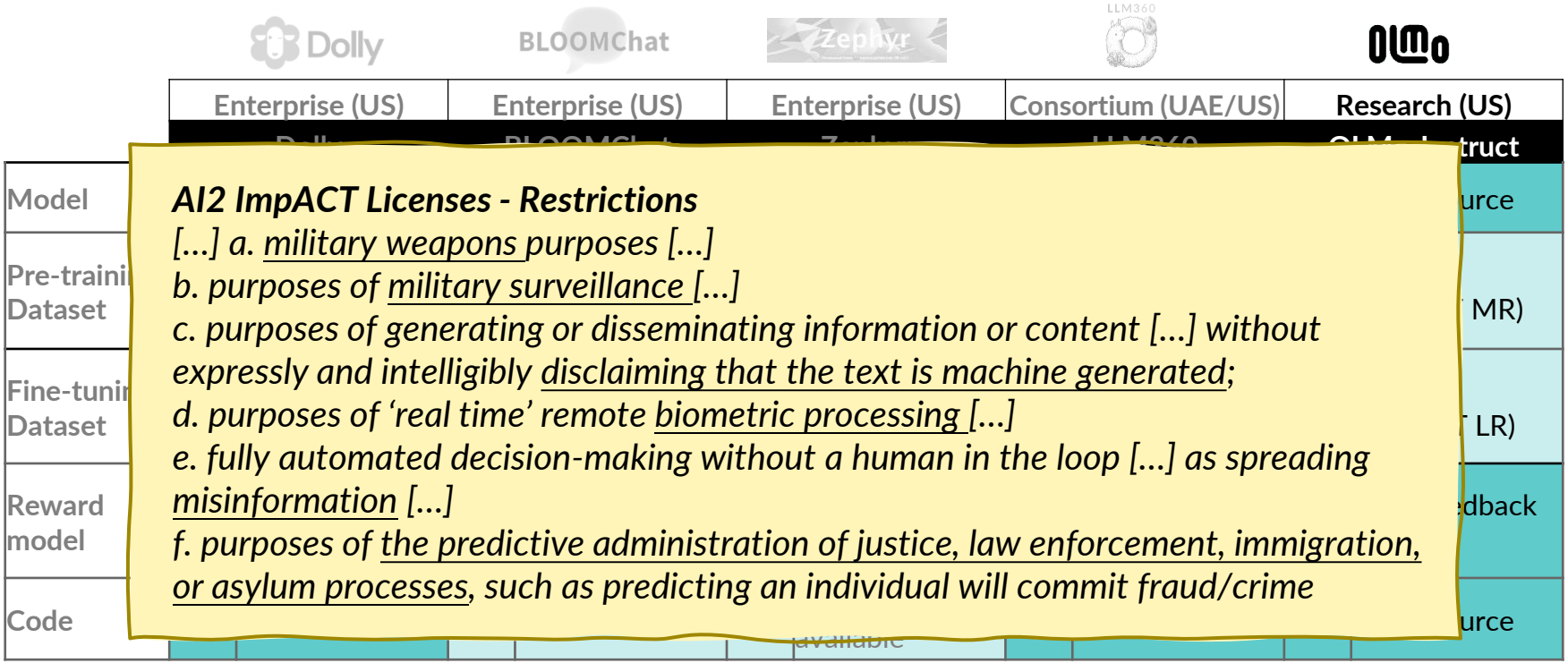
Just like with Open Source, revisiting ***Open*** **in Cloud era**

# Collaborative **fine-tuned** LLMs

| | Dolly | | BLOOMChat | | Zephyr | | LLM360 | | OLMo-Instruct | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Enterprise (US) | | Enterprise (US) | | Enterprise (US) | | Consortium (UAE/US) | | Research (US) | |
| **Model** | 4 | Based on GPT-J | 3 | Based on BLOOM | 4 | Based on Mistral | 4 | Open source | 4 | Open source |
| **Pre-training Dataset** | 3 | Based on GPT-J | 3 | Based on BLOOM | 0 | Based on Mistral | 4 | RedPajama, Falcon, StarCoder | 3 | Dolma (ImpACT MR) |
| **Fine-tuning Dataset** | 4 | Access and reuse without restriction | 4 | Dolly and LAION | 2 | Research use only (OpenAI) | 2 | Research use only (OpenAI) | 3 | Tülu 2 (IMPACT LR) |
| **Reward model** | 0 | No public information available | 0 | No public information available | 3 | Paper and code examples | 0 | No public information available | 4 | UltraFeedback (MIT) |
| **Code** | 4 | Open source | 3 | OpenRAIL | 3 | Example code available | 4 | Open source | 4 | Open source |

**Impact** of foundational models or pre-training datasets

# Collaborative **fine-tuned** LLMs




**Responsible AI** contradicts Open Source Definition

# Other aspects of GenAI's **Linux Moment**

**Democratize and Decentralize (re)use and innovation**

## Collaborative Tools & Ecosystems







Notebooks
Communities
New Business Models

## Hardware Optimization







AI Chips
Quantization
Decentralization

## Opensource Tools & Frameworks







*Do One Thing Well*
Interoperable Standards
Beyond Python

# Key **takeaways**



- **Closed APIs → Open Weights → Free AI (as in freedom)**
  - Datasets and upstream transitivity
  - Competitive clauses
  - Responsible AI restrictions
- **Open Research → Competitive Market → Coopetitive Ecosystem**
  - Openness fosters reuse and collaboration
  - Collaboration brings commoditization and innovation

**Just like Open Source!**



# Thank you

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